

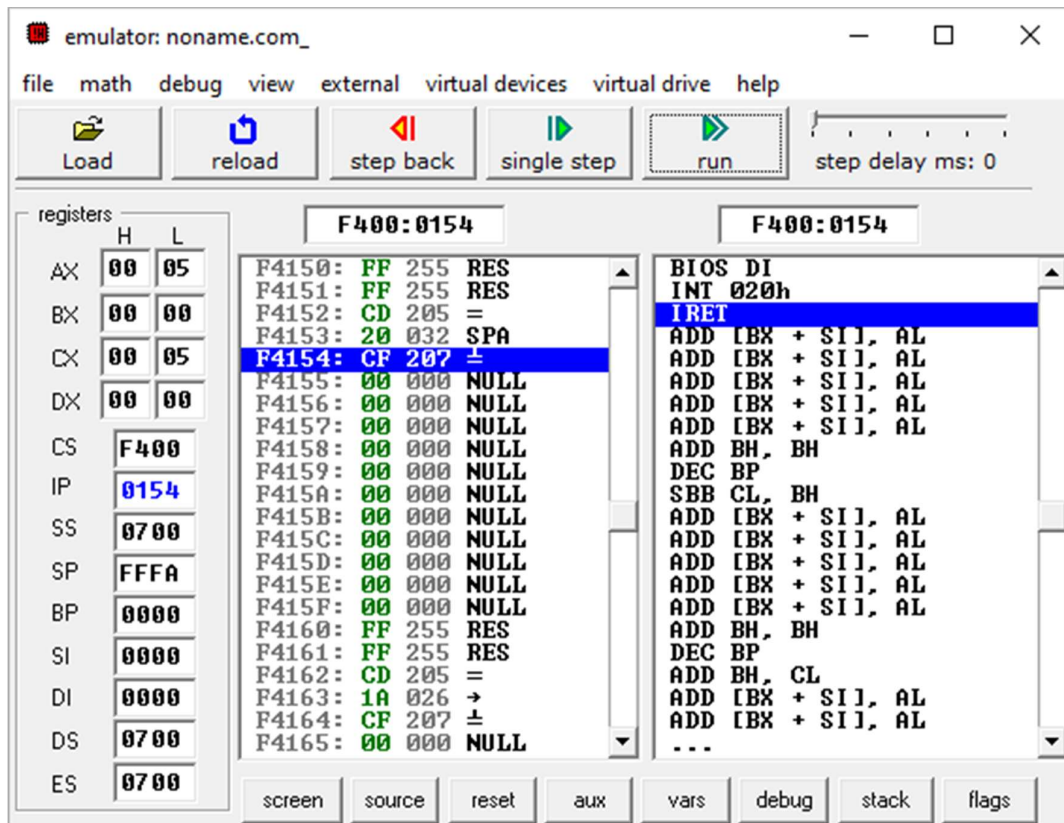
COAL LAB #5

Exercise#1

PROGRAM 1:

```
org 100h
MOV AL, 4
INC AL
RET
```

OUTPUT:



PROGRAM 2:

```
org 100h
MOV AL, 255      ; AL = 0FFh (255 or -1)
DEC AL          ; AL = 0FEh (254 or -2)
RET
```

OUTPUT:

emulator: noname.com_

file math debug view external virtual devices virtual drive help

Load reload step back single step run step delay ms: 0

registers

	H	L
AX	00	FE
BX	00	00
CX	00	05
DX	00	00
CS	F400	
IP	0154	
SS	0700	
SP	FFFA	
BP	0000	
SI	0000	
DI	0000	
DS	0700	
ES	0700	

F400:0154

F4150:	FF	255	RES
F4151:	FF	255	RES
F4152:	CD	205	=
F4153:	20	032	SPA
F4154:	CF	207	±
F4155:	00	000	NULL
F4156:	00	000	NULL
F4157:	00	000	NULL
F4158:	00	000	NULL
F4159:	00	000	NULL
F415A:	00	000	NULL
F415B:	00	000	NULL
F415C:	00	000	NULL
F415D:	00	000	NULL
F415E:	00	000	NULL
F415F:	00	000	NULL
F4160:	FF	255	RES
F4161:	FF	255	RES
F4162:	CD	205	=
F4163:	1A	026	→
F4164:	CF	207	±
F4165:	00	000	NULL

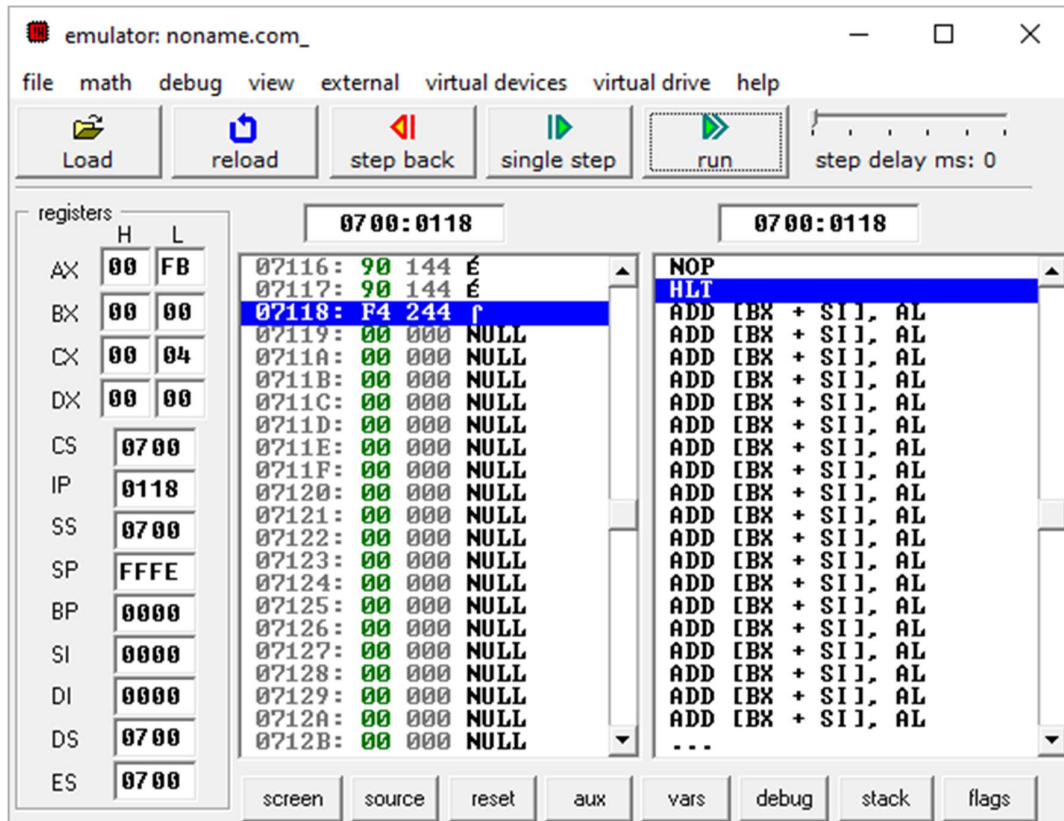
F400:0154

BIOS DI
INT 020h
IRET
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD BH, BH
DEC BP
SBB CL, BH
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD BH, BH
DEC BP
ADD BH, CL
ADD [BX + SI], AL
ADD [BX + SI], AL
...

screen source reset aux vars debug stack flags

PROGRAM 3:

```
org 100h
MOV AL, 5 ; AL = 05h
NEG AL ; AL = 0FBh (-5)
```

OUTPUT:

PROGRAM 4:

```

Org 100h
MOV AL, 5    ; AL=5
ADD AL, -3   ; AL=2
RET

```

OUTPUT:

The screenshot shows an x86 emulator window titled "emulator: noname.com_". The interface includes a menu bar (file, math, debug, view, external, virtual devices, virtual drive, help) and a toolbar with buttons for Load, reload, step back, single step, run, and a step delay slider set to 0 ms.

The "registers" panel on the left displays the following values:

	H	L
AX	00	02
BX	00	00
CX	00	05
DX	00	00
CS	F400	
IP	0154	
SS	0700	
SP	FFFA	
BP	0000	
SI	0000	
DI	0000	
DS	0700	
ES	0700	

The main memory window shows two panels, both titled "F400:0154".

The left panel displays memory contents:

Address	Hex	Dec	Comment
F4150:	FF	255	RES
F4151:	FF	255	RES
F4152:	CD	205	=
F4153:	20	032	SPA
F4154:	CF	207	±
F4155:	00	000	NULL
F4156:	00	000	NULL
F4157:	00	000	NULL
F4158:	00	000	NULL
F4159:	00	000	NULL
F415A:	00	000	NULL
F415B:	00	000	NULL
F415C:	00	000	NULL
F415D:	00	000	NULL
F415E:	00	000	NULL
F415F:	00	000	NULL
F4160:	FF	255	RES
F4161:	FF	255	RES
F4162:	CD	205	=
F4163:	1A	026	→
F4164:	CF	207	±
F4165:	00	000	NULL

The right panel displays BIOS code:

```

BIOS DI
INT 020h
IRET
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD BH, BH
DEC BP
SBB CL, BH
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD BH, BH
DEC BP
ADD BH, CL
ADD [BX + SI], AL
ADD [BX + SI], AL
...

```

At the bottom of the window are buttons for screen, source, reset, aux, vars, debug, stack, and flags.

PROGRAM 5:

```

org 100h
MOV AL, 5
SUB AL, 1 ; AL=4
RET

```

OUTPUT:

emulator: noname.com_

file math debug view external virtual devices virtual drive help

Load reload step back single step run step delay ms: 0

registers

	H	L
AX	00	04
BX	00	00
CX	00	05
DX	00	00
CS	F400	
IP	0154	
SS	0700	
SP	FFFA	
BP	0000	
SI	0000	
DI	0000	
DS	0700	
ES	0700	

F400:0154

F4150:	FF	255	RES
F4151:	FF	255	RES
F4152:	CD	205	=
F4153:	20	032	SPA
F4154:	CF	207	±
F4155:	00	000	NULL
F4156:	00	000	NULL
F4157:	00	000	NULL
F4158:	00	000	NULL
F4159:	00	000	NULL
F415A:	00	000	NULL
F415B:	00	000	NULL
F415C:	00	000	NULL
F415D:	00	000	NULL
F415E:	00	000	NULL
F415F:	00	000	NULL
F4160:	FF	255	RES
F4161:	FF	255	RES
F4162:	CD	205	=
F4163:	1A	026	→
F4164:	CF	207	±
F4165:	00	000	NULL

F400:0154

BIOS DI
INT 020h
IRET
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD BH, BH
DEC BP
SBB CL, BH
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD BH, BH
DEC BP
ADD BH, CL
ADD [BX + SI], AL
ADD [BX + SI], AL
...

screen source reset aux vars debug stack flags

PROGRAM 6:

```
Org 100h
MOV AL, 200      ; AL = 0C8h
MOV BL, 4
MUL BL          ; AX = 0320h (800)
RET
```

OUTPUT:

The screenshot shows an emulator window titled "emulator: noname.com_". The menu bar includes "file", "math", "debug", "view", "external", "virtual devices", "virtual drive", and "help". The toolbar contains buttons for "Load", "reload", "step back", "single step", "run", and a "step delay ms: 0" slider.

The "registers" panel on the left displays the following values:

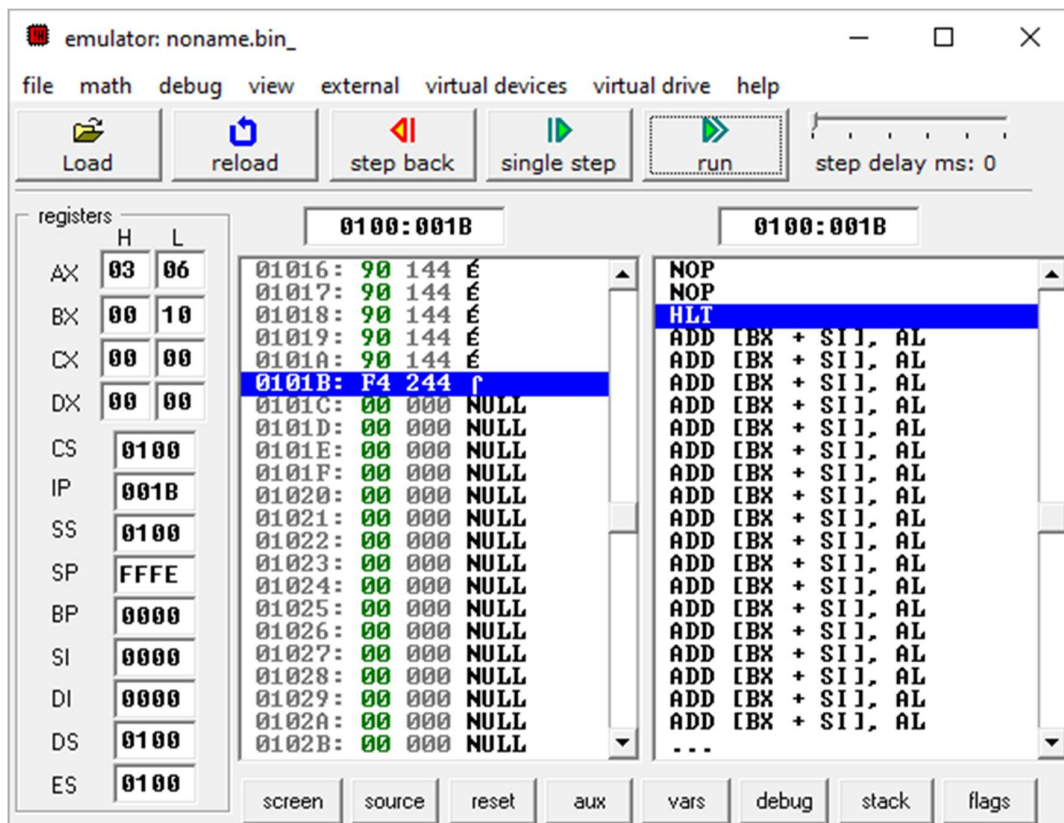
	H	L
AX	00	00
BX	00	00
CX	00	07
DX	00	00
CS	0700	
IP	0100	
SS	0700	
SP	FFFE	
BP	0000	
SI	0000	
DI	0000	
DS	0700	
ES	0700	

The main window displays two assembly code windows. The left window shows memory addresses from 07100 to 07115 with their corresponding hex values and assembly instructions. The right window shows the assembly instructions being executed.

Address	Hex	Instruction
07100	B0 176	MOV AL, 0C8h
07101	C8 200	MOV BL, 04h
07102	B3 179	MUL BL
07103	04 004	RET
07104	F6 246	NOP
07105	E3 227	NOP
07106	C3 195	NOP
07107	90 144	NOP
07108	90 144	NOP
07109	90 144	NOP
0710A	90 144	NOP
0710B	90 144	NOP
0710C	90 144	NOP
0710D	90 144	NOP
0710E	90 144	NOP
0710F	90 144	NOP
07110	90 144	NOP
07111	90 144	NOP
07112	90 144	NOP
07113	90 144	NOP
07114	90 144	NOP
07115	90 144	NOP

The bottom of the window has tabs for "screen", "source", "reset", "aux", "vars", "debug", "stack", and "flags".

Exercise #2



AH	AL
03	06

Exercise #3

CODE:

```

org 100h
mov ax, 5
mov bx, 3
mul bx
mov var1, ax
ret
var1 dw 0

```

OUTPUT:

emulator: noname.com_

file math debug view external virtual devices virtual drive help

Load reload step back single step run step delay ms: 0

registers		F400:0154		F400:0154	
	H	L			
AX	00	0F	F4150: FF 255 RES		BIOS DI
BX	00	03	F4151: FF 255 RES		INT 020h
CX	00	0E	F4152: CD 205 =		I RET
DX	00	00	F4153: 20 032 SPA		ADD [BX + SI], AL
CS	F400		F4154: CF 207 ±		ADD [BX + SI], AL
IP	0154		F4155: 00 000 NULL		ADD [BX + SI], AL
SS	0700		F4156: 00 000 NULL		ADD [BX + SI], AL
SP	FFFA		F4157: 00 000 NULL		ADD [BX + SI], AL
BP	0000		F4158: 00 000 NULL		ADD [BX + SI], AL
SI	0000		F4159: 00 000 NULL		ADD BH, BH
DI	0000		F415A: 00 000 NULL		DEC BP
DS	0700		F415B: 00 000 NULL		SBB CL, BH
ES	0700		F415C: 00 000 NULL		ADD [BX + SI], AL
			F415D: 00 000 NULL		ADD [BX + SI], AL
			F415E: 00 000 NULL		ADD [BX + SI], AL
			F415F: 00 000 NULL		ADD [BX + SI], AL
			F4160: FF 255 RES		ADD [BX + SI], AL
			F4161: FF 255 RES		ADD BH, BH
			F4162: CD 205 =		DEC BP
			F4163: 1A 026 →		ADD BH, CL
			F4164: CF 207 ±		ADD [BX + SI], AL
			F4165: 00 000 NULL		ADD [BX + SI], AL
					...

screen source reset aux vars debug stack flags

Exercise #4

CODE:

```

org 100h
mov ax, 276
mov bl, 10
div bl
mov var1, ax
ret
var1 dw 0

```

OUTPUT:

The screenshot shows an emulator window titled "emulator: noname.com_". The interface includes a menu bar (file, math, debug, view, external, virtual devices, virtual drive, help) and a toolbar with buttons for Load, reload, step back, single step, run, and a step delay slider set to 0 ms.

The "registers" panel on the left displays the following values:

	H	L
AX	06	1B
BX	00	0A
CX	00	0D
DX	00	00
CS	F400	
IP	0154	
SS	0700	
SP	FFFA	
BP	0000	
SI	0000	
DI	0000	
DS	0700	
ES	0700	

The main memory window shows two panels, both titled "F400:0154".

The left panel displays memory contents:

Address	Hex	Dec	Comment
F4150:	FF	255	RES
F4151:	FF	255	RES
F4152:	CD	205	=
F4153:	20	032	SPA
F4154:	CF	207	±
F4155:	00	000	NULL
F4156:	00	000	NULL
F4157:	00	000	NULL
F4158:	00	000	NULL
F4159:	00	000	NULL
F415A:	00	000	NULL
F415B:	00	000	NULL
F415C:	00	000	NULL
F415D:	00	000	NULL
F415E:	00	000	NULL
F415F:	00	000	NULL
F4160:	FF	255	RES
F4161:	FF	255	RES
F4162:	CD	205	=
F4163:	1A	026	→
F4164:	CF	207	±
F4165:	00	000	NULL

The right panel displays BIOS code:

Address	Hex	Dec	Comment
BIOS DI			
INT 020h			
I RET			
ADD [BX + SI], AL			
ADD [BX + SI], AL			
ADD [BX + SI], AL			
ADD [BX + SI], AL			
ADD [BX + SI], AL			
ADD BH, BH			
DEC BP			
SBB CL, BH			
ADD [BX + SI], AL			
ADD [BX + SI], AL			
ADD [BX + SI], AL			
ADD [BX + SI], AL			
ADD [BX + SI], AL			
ADD [BX + SI], AL			
ADD BH, BH			
DEC BP			
ADD BH, CL			
ADD [BX + SI], AL			
ADD [BX + SI], AL			
...			

At the bottom of the window are buttons for screen, source, reset, aux, vars, debug, stack, and flags.